| | | | |
|---|---|---|---|
| **Fiscal Year:** | FY 2005 | **Task Last Updated:** | FY 06/22/2006 |
| **PI Name:** | Hasenstein, Karl Ph.D. | | |
| **Project Title:** | Application of Physical and Biological Techniques in the Study of the Gravisensing and Response System of Plants | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | SPACE BIOLOGY | | |
| **Program/Discipline--Element/Subdiscipline:** | SPACE BIOLOGY--Cellular and molecular biology | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | hasenstein@louisiana.edu | **Fax:** | FY 337-482-5834 |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 337-482-6750 |
| **Organization Name:** | University of Louisiana - Lafayette | | |
| **PI Address 1:** | PO Box 42451 | | |
| **PI Address 2:** | 300 E. St. Mary Boulevard | | |
| **PI Web Page:** | http://www.ucs.louisiana.edu/ | | |
| **City:** | Lafayette | **State:** | LA |
| **Zip Code:** | 70504-2451 | **Congressional District:** | 3 |
| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 95-OLMSA-01 |
| **Start Date:** | 03/25/1996 | **End Date:** | 09/30/2006 |
| **No. of Post Docs:** | 2 | **No. of PhD Degrees:** | 2 |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | 1 |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | 3 |
| **No. of Bachelor's Candidates:** | 1 | **Monitoring Center:** | NASA KSC |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | Shuttle | | |
| **Flight Assignment:** | STS-107 | | |
| **Key Personnel Changes/Previous PI:** | n/a | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | NAG10-0190 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| | This proposal combines ground-based research with research on the Shuttle. It was proposed to establish and maintain directional growth of plant seedlings and eventually mature plants under microgravity conditions using high gradient magnetic fields (HGMF). Plant material from these experiments will be used for investigations on the cytoskeleton of plants grown in micro-gravity. Space research will establish threshold levels of the magnetic field strength required for the establishment and maintenance of directional growth in plants during germination and growth under microgravity conditions. The experiments allow to investigate whether the force exerted by amyloplasts or their position controls growth. HGMF exert a directional, repulsive force on diamagnetic substances such as the starch in amyloplasts. Preliminary data show that amyloplast in root cap cells move along the gradient of magnetic field. Space experimentation is necessary to establish whether the theoretically predicted threshold value for magnetic field is capable | | |

| | |
|---|---|
| **Task Description:** | to direct and maintain directional growth in the absence of a gravity field, i.e. substitute for the gravity vector. Experiments on the whole plant level will measure rate and control of root and shoot elongation. Studies on the cellular level will examine the arrangement /behavior of cortical microtubul(MTs) to test whether differences in the cytoskeletal organization (and consequently deposition of cell wall material) exist in space compared to 1 g controls. We intend to test whether the cytoskeletal changes in curving, space grown plants are comparable to those found in graviresponding plants at 1 g. This will provide insight in the fundamental mechanism of how the graviresponse system of plants is organized and operates. The experiments will provide answers as to whether plant growth can be controlled by only affecting amyloplasts in the growth tip or whether the whole plant body needs and responds to mechanical stimulation. The objective of the experiment “magnetophoretic induction of curvature in roots” (MICRO) was the determination of effects of magnetophoretically displaced, starch-filled amyloplasts on root curvature in the microgravity environment on STS-107. The Shuttle Experiment was designed to take advantage of the magnetic properties of amyloplasts in gravity-sensing cells (statocytes), which respond to the intracellular displacement of these organelles. The scientific objectives included measuring the force that plants perceive, investigating the growth/curvature pattern as roots pass by the force-inducing high-gradient magnetic field (HGMF), the determination if amyloplasts are the gravisensing/curvature-inducing structures, and lastly a study of the effects of HGMF and microgravity on the plant cytoskeleton. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The application of high gradient magnetic fields to investigate the gravity sensing mechanism of plants has implications on two levels. First, the research utilizes and improves a novel mechanism of intracellular displacement of starch filled amyloplasts and the resulting growth response of plants. Secondly, the growth response is likely to be tightly linked to the perception of a stimulus analogous to gravity. Therefore the research addresses the larger problem of studying the signal perception/response mechanism in plants. Such studies promote our understanding of plant growth regulation. In particular, the high gradient magnetic field-dependent growth response will elucidate the change in elongation growth and thus directional and differential deposition of cell wall material. In addition, as with all basic research, an improved understanding of basic growth phenomena will benefit the average citizen because of a better understanding for improving growth, biomass production on earth, and the bio-mechanic properties of plants. |
| **Task Progress:** | Intended research could not be completed because of the Columbia space shuttle accident. However, the discovery that microgravity-grown roots appeared to be more sensitive to magnetophoretic stimulation prompted experiments under short-term microgravity during parabolic flights on board NASA's C9 aircraft. The data confirmed that magnetophoresis is a suitable technique for intracellular amyloplast displacement. |
| **Bibliography Type:** | Description: (Last Updated: 01/20/2026) |
| **Articles in Peer-reviewed Journals** | Hasenstein KH, Scherp P, Ma Z. "Gravisensing in flax roots--results from STS 107." Advances in Space Research 2005;36(7):1189-95. http://dx.doi.org/10.1016/j.asr.2005.01.007 , Jul-2005 |
| **Articles in Peer-reviewed Journals** | Ma Z, Hasenstein KH. "The onset of gravisensitivity in the embryonic root of flax." Plant Physiol. 2006 Jan;140(1):159-66. PMID: 16377740 , Jan-2006 |
| **Articles in Peer-reviewed Journals** | Ma Z, Hasenstein KH. "Noise amplification of plant gravisensing." Advances in Space Research. 2007;39(7):1119-26. https://doi.org/10.1016/j.asr.2006.12.005 , Mar-2007 |